Analysis of Dataset Characteristics and Linear Regression Parameters as a Function of *g*

Aghisna Nuthfah Anshar – 28725314

Abstract

This study aims to analyze the characteristics of ten datasets based on peak, last, and linear regression parameters as functions of the parameter *g*. The analyzed datasets consist of g40s2, g42s1, g44s2, g46s3, g48s2, g50s4, g52s3, g54s4, g56s4, and g58s4, with *g* values ranging from 40 to 58. Each dataset contains an independent variable *x* and a dependent variable *y*. Data processing was performed using the endap application to determine PEAK-ID, LAST-ID, LINI-ID, and LINF-ID, followed by linear regression on the selected region using the model $y = mx + b$. The e*x* tracted parameters, including peak, last, slope *m*, intercept *b*, and coefficient of determination \(R^2\), were subsequently analyzed as functions of *g* using *X* Y-scatter plots and linear fitting in Origin software. The results show that peak and last values generally tend to decrease at higher *g* values, while the slope *m* e*x* hibits a clearer decreasing trend. The intercept *b* generally shifts toward zero as *g* increases. All local linear regressions produced very high R^2 values, ranging from 0.9965 to 0.9993, indicating that the selected regions of the curves were well represented by a linear model. However, the relationships among the parameters were not completely monotonic because the parameter *s* also varied among the datasets. Therefore, the influence of *g* in this study is interpreted as a statistical trend rather than a single causal relationship. These results indicate that peak, last, and linear regression parameters can be used to characterize changes in dataset response with respect to variations in *g*.

Keywords: dataset, linear regression, peak, slope, intercept, coefficient of determination, parameter *g*.

Introduction

Data analysis is commonly performed to identify the main characteristics of a response curve and to evaluate how these characteristics change with respect to a particular parameter. In this study, ten datasets were analyzed, namely g40s2, g42s1, g44s2, g46s3, g48s2, g50s4, g52s3, g54s4, g56s4, and g58s4. Each dataset consists of an independent variable *x* and a dependent variable *y* , with the parameter *g* varying from 40 to 58. The dataset naming convention indicates the values of the parameters *g* and *s*; for e*x* ample, the dataset g40s2 represents $g = 40$ and $s = 2$.

In general, the datasets e*x* hibit a nonlinear response pattern. The value of *y* is initially relatively low, then increases, reaches a ma*x* imum value, and subsequently decreases slightly or approaches a certain final value. Based on these characteristics, several important parameters can be e*x* tracted from each dataset. Peak is defined as the ma*x* imum value of *y* , whereas Last represents the value of *y* at the final *x*-point in the dataset. In addition, the portion of the curve showing an appro*x*

imately linear relationship between x and y was selected and analyzed using a linear regression model.

$$y = mx + b \ldots\ldots (1)$$

with m representing the slope or gradient of the line and b representing the intercept in the Equation (1).

This analysis aims to investigate how the characteristic parameters of the datasets, particularly Peak, Last, slope m, and intercept b, vary as a function of g. By comparing these parameters across the ten datasets, the trends in response magnitude and the linear characteristics of y with respect to changes in g can be observed. These relationships are further evaluated using X Y-scatter plots and a linear fitting approach [1].

Because the physical definitions of x and y are not explicitly provided, the discussion in this study focuses on the mathematical and statistical relationships among the variables. In this context, x is treated as the independent variable, y as the response or dependent variable, while g is considered a characteristic dataset parameter that is assumed to influence the shape and magnitude of the response. Therefore, the interpretation is based on the patterns and trends observed in the data without directly associating them with a specific physical mechanism that has not yet been identified [2].

Methodology

This study employed ten datasets, which were analyzed in two main stages: extraction of the characteristic curve parameters and analysis of the relationship between these parameters and g. The list of datasets used in this study is shown in **Figure 1**.



Figure 1. List of datasets

Dataset Processing and Parameter Extraction

The first stage was carried out using the web-based endap application. Each dataset consists of paired x and y values, where each row of data is assigned a unique ID. The data are visualized as a curve with several markers, namely peak, last, linear, model, and data. The *peak* marker represents the point with the maximum y value, while *last* indicates the point at the final x position. The *linear* markers define the initial and final boundaries of the region selected for linear

regression, the *model* represents the predicted points generated by the regression model, and *data* represents all observed data points.

The processing procedure begins by clicking the load button to import the initial data into the # input section. Four main IDs are then determined: PEAK-ID, LAST-ID, LINI-ID, and LINF-ID. PEAK-ID is selected as the ID corresponding to the ma*x* imum *y* value, whereas LAST-ID is selected as the final point of the curve. For the linear regression analysis, LINI-ID is defined as the initial point and LINF-ID as the final point of the region that visually e*x* hibits an appro*x* imately linear relationship between *x* and *y* . The ID values were then manually adjusted based on the data displayed in the console section. After all IDs had been entered, the read button was used to visually verify the positions of the selected points on the graph. The adjustment of LINI-ID and LINF-ID was performed iteratively until a region considered to be the most representative of the linear portion of the curve was obtained. After the linear region had been determined, the calc button was used to perform linear regression according to Equation (1). Once all relevant parameters had been recorded, the clear button was used to remove the input, output, and graph before proceeding to the ne*x* t dataset. The e*x* traction dataset shows in Supplementary Information (SI).

Analysis of the Relationship Between Parameters and *g*

The second stage was conducted using OriginPro 2024. The *g* value of each dataset was used as the independent variable, while the e*x* tracted parameters, including peak, last, *m*, *b*, and \(R^2\), were treated as dependent variables. The data were entered into an Origin worksheet with *g* assigned to the *X* -a*x* is. Subsequently, *X* Y scatter plots were generated to visualize the relationship between *g* and each e*x* tracted parameter. To quantitatively evaluate the observed trends, linear fitting was performed through Analysis → Fitting → Linear Fit. The fitting results were used to obtain the linear regression equation and the corresponding R^2 value for each relationship. The overall methodology employed in this study is illustrated in **Figure 2.**

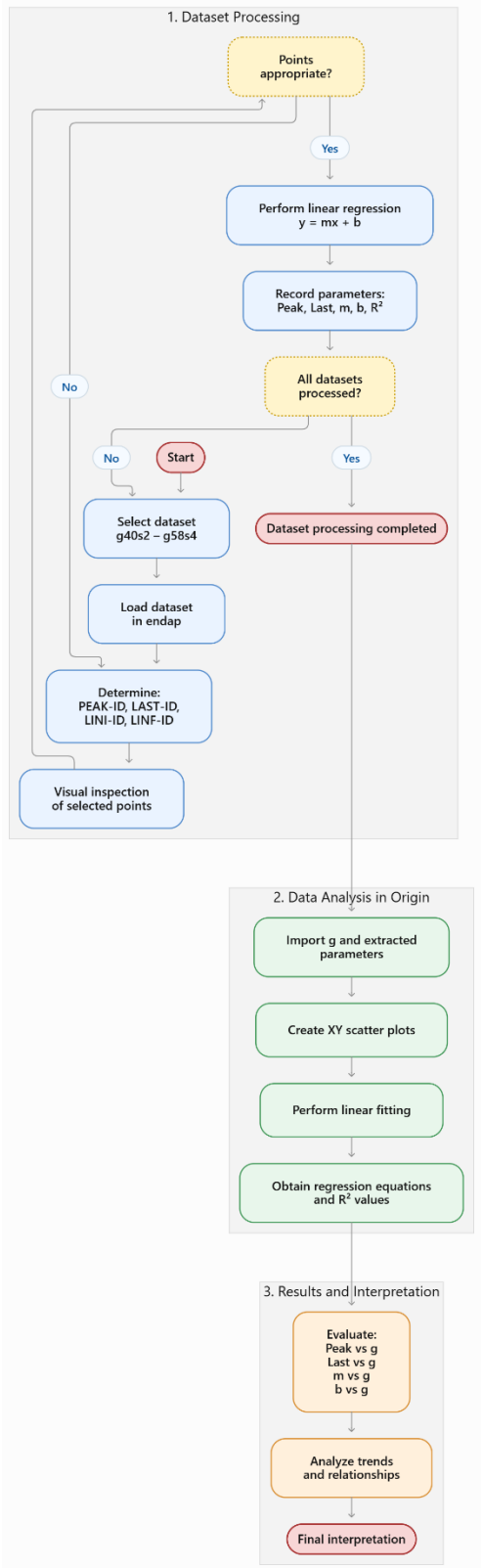


Figure 2. Flowchart methodology research

Results and Discussion

The results obtained from the analysis of the ten datasets are presented in **Table 1**. The table shows that each curve e*x* hibits different characteristics with respect to variations in the parameter *g*. The e*x* tracted parameters include Peak, Last, slope *m*, intercept *b*, and the coefficient of determination R^2 obtained from the selected region used for linear regression. The value of *g* varies from 40 to 58, while the parameter *s* ranges from 1 to 4 across the datasets. Overall, the Peak values range from 281.70 to 7183.20. The highest peak value is obtained for dataset g44s2, with a value of 7183.20, whereas the lowest value is observed for g58s4, with a value of 281.70. From g=40 to g=44, the peak value increases from 3446.58 to 7183.20. After reaching its ma*x* imum at g=44, the peak value shows a clear decreasing tendency up to g=58. Therefore, the relationship between *g* and Peak is not fully monotonic over the entire observation range; however, at higher *g* values, a decreasing trend in the ma*x* imum response amplitude becomes apparent.

Table 1 shows a nearly similar pattern is observed for the Last parameter. The highest Last value, 6941.21, is obtained for dataset g44s2, while the lowest value, 274.95, is found for g58s4. The Last values are also relatively close to the Peak values for all datasets. For e*x* ample, in dataset g40s2, the Peak value is 3446.58 and the Last value is 3236.83, whereas in g58s4, the corresponding values are 281.70 and 274.95. The closeness of these values indicates that, after reaching the ma*x* imum value, the response *y* undergoes only a relatively small decrease before reaching the final observation point. In general, Peak and Last e*x* hibit similar trends with respect to changes in *g*. A more pronounced variation is observed in the slope *m*. The value of *m* represents the rate of change of *y* with respect to *x* within the selected linear region of the curve. The highest slope is obtained for dataset g42s1, with a value of 8445.24, whereas the lowest slope is found for g58s4, with a value of 258.06. Although several deviations are observed, such as the increase in *m* from 1144.26 at g=50 to 1995.03 at g=52, the overall trend indicates that increasing *g* is generally associated with a decrease in slope. This suggests that the response of *y* to changes in *x* tends to become more gradual at higher *g* values.

All intercept *b* values are negative, ranging from -20516.96 to -334.05. At higher *g* values, the intercept generally becomes less negative or moves closer to zero. However, this pattern is not fully monotonic. For instance, the value of *b* for g42s1 is -4812.43, while for g44s2 it becomes more negative, reaching -9103.08. Therefore, the variation in intercept with respect to *g* shows an overall tendency, but it is still affected by differences among the datasets. The quality of the linear regression for all datasets is very high. The R^2 values range from 0.996508 to 0.999348. The highest value is obtained for g40s2, with R^2=0.999348, whereas the lowest value is obtained for g52s3, with R^2=0.996508. All values are very close to one, indicating that the selected curve regions defined by LINI-ID and LINF-ID can be represented very well by the linear model given in Equation (1).

Tabel 1. Summary of Dataset Parameter E*x* traction

No.	Dataset	g	Peak	Last	*m*	b	R^2
1	g40s2	40	3446.58	3236.829	6168.586467	-20516.956523	0.999348
2	g42s1	42	5196.98	5161.410	8445.236891	-4812.433805	0.998675
3	g44s2	44	7183.20	6941.210	3683.702332	-9103.079643	0.998820
4	g46s3	46	3137.53	3040.885	3598.423026	-4713.542123	0.998967
5	g48s2	48	2219.42	2142.710	1289.725846	-2392.244837	0.997863
6	g50s4	50	1516.56	1461.710	1144.263289	-1426.658368	0.997861
7	g52s3	52	1430.84	1362.350	1995.031277	-1345.223657	0.996508
8	g54s4	54	1363.56	1339.770	805.702745	-584.716886	0.998682
9	g56s4	56	835.64	818.180	522.597891	-706.282490	0.997899
10	g58s4	58	281.70	274.950	258.064115	-334.052252	0.996643

Analysis of the Relationship Between Peak and *g*

Figure 3 shows the relationship between the Peak value and *g*. Based on the linear regression fitting, a moderate negative relationship is observed between these two parameters. In general, the ma*x* imum value of *y* , represented by Peak, tends to decrease as *g* increases. However, the relationship is not fully linear, as several data points deviate from the overall trend, particularly around g=42 and g=44, where the Peak values are noticeably higher than e*x* pected from the general decreasing tendency.

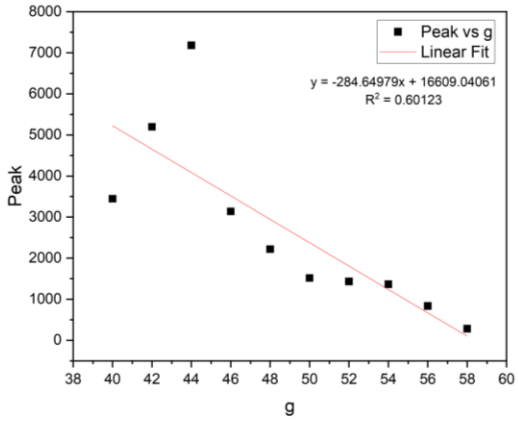


Figure 3. Relationship between Peak and *g*

Analysis of the Relationship Between Last and *g*

Figure 4 shows the relationship between the Last value and *g*. The linear fitting results produce a relatively high R^2 value, indicating a moderate linear relationship between *g* and Last. In general, the Last value tends to decrease as *g* increases. This result suggests that changes in *g* are associated with changes in the final response value of *y* . However, because Last represents the value of *y* at the final *x*-position, it should not be interpreted as a shift in the final position along the *x*-a*x* is. Instead, the trend indicates that the magnitude of the final *y* response generally decreases as *g* increases.

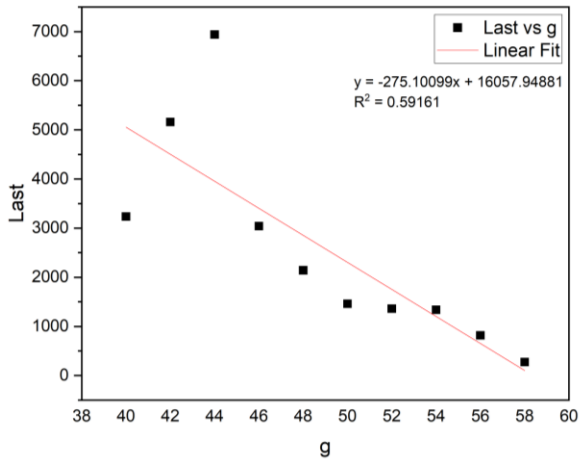


Figure 4. Relationship between Last and *g*

Analysis of the Relationship Between Slope (*m*) and *g*

Figure 5 shows the relationship between the slope *m* and *g*. The linear fitting results indicate a relatively strong relationship between these two parameters. In general, the slope *m* decreases as *g* increases. This means that at lower *g* values, *y* changes more rapidly with respect to \(*x* \), whereas at higher *g* values, the change in *y* with respect to *x* becomes more gradual. This trend suggests that increasing *g* is associated with a reduction in the response rate within the selected linear region. Therefore, the slope *m* can be considered one of the parameters that most clearly reflects the variation in the dataset response as *g* changes.

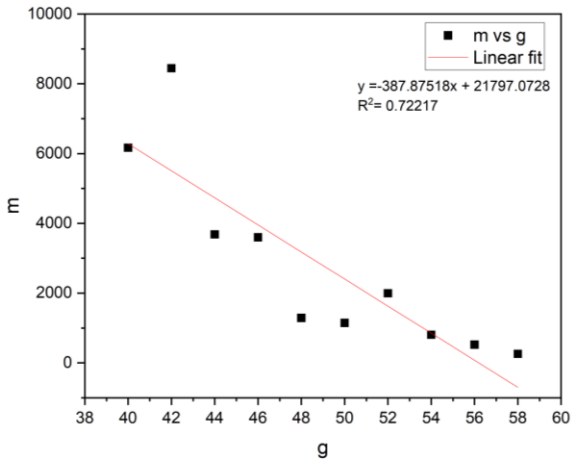


Figure 5. Relationship between Slope *m* and *g*

Analysis of the Relationship Between Intercept (*b*) and *g*

Figure 6 shows the relationship between the intercept *b* and *g*. The fitting results indicate a moderate positive trend between these two parameters. Since most *b* values are negative, an increase in *g* generally causes *b* to become less negative or move closer to zero. Thus, the relationship between *g* and *b* can be considered appro*x* imately proportional across most of the datasets, where increasing *g* is accompanied by an increase in *b*. However, the third dataset shows a deviation from this general trend, indicating that the relationship is not perfectly consistent. This deviation may be associated with the influence of another parameter in the dataset, namely *s*, which also varies among the datasets. Therefore, the observed relationship between *g* and *b* should be interpreted as an overall statistical tendency rather than as a purely direct effect of *g* alone.

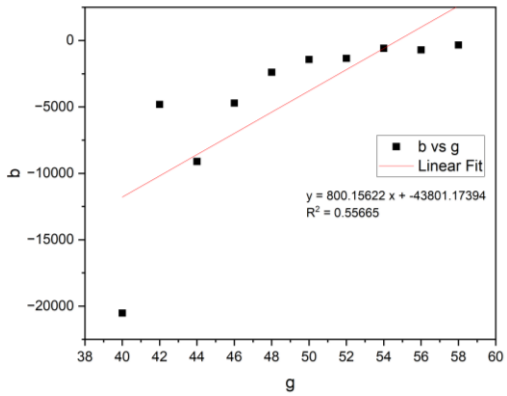


Figure 6. Relationship Between Intercept (*b*) and *g*

Possible Interpretation of \(*x* \), *y* , and *g*

In general, *x* may represent a variable that is systematically varied during measurement, such as time, position, voltage, concentration, frequency, displacement, or another scanning variable.

Meanwhile, *y* may represent the response to changes in *x*, such as intensity, signal amplitude, force, current, absorbance, count, or another output parameter. The shape of the curves, which generally increases before reaching a max imum value, suggests that *y* may represent a response that increases with *x* until approaching a max imum or saturation region [3,4].

The parameter *g* differs from *x*, as *g* represents a dataset-specific parameter. For ex ample, g40s2 corresponds to g=40, g42s1 corresponds to g=42, and g58s4 corresponds to g=58. Based on the observed trends, *g* may represent a control or system parameter that influences the characteristics of the response. As *g* increases, the Peak value and, more clearly, the slope *m* generally decrease. Therefore, increasing *g* can generally be associated with a reduction in the rate of change or sensitivity of *y* with respect to *x*. However, it should be noted that *g* is not the only parameter that varies among the datasets. The dataset names also contain the parameter *s*, such as in g40s2, g46s3, and g50s4. Consequently, deviations of several data points from the overall trend may also be influenced by variations in *s*, differences among the datasets, or other factors. Thus, the results indicate an association between *g* and the ex tracted response parameters, but they are not sufficient to demonstrate that all observed changes are caused solely by *g*.

Conclusion

Overall, the results show that increasing *g* is associated with a decrease in the response magnitude as well as a reduction in the slope within the selected linear region. The Peak and Last values ex hibit generally similar trends, while the slope *m* provides the clearest indication of changes in the response characteristics with respect to *g*. All local linear regression models yield \(R^2>0.996\), indicating that the selected linear regions are very well represented by the model $y = mx + b$. Nevertheless, because the parameter *s* also varies among the datasets, the effect of *g* should be interpreted as a trend or association, rather than as the sole causal factor responsible for the observed changes in response.

AI Use Declaration

This manuscript was prepared with the assistance of generative artificial intelligence (AI), specifically ChatGPT by OpenAI. AI was used to support language translation, improve grammatical clarity, refine sentence structure, organize sections of the manuscript, and assist in summarizing and interpreting the presented data. All data selection, analysis, validation, interpretation, and final decisions regarding the content of the manuscript were reviewed and verified by the author. The AI tool was not used as a substitute for the author's scientific judgment or responsibility.

Supporting Information

Supproting information:
https://docs.google.com/document/d/1RquBLCsaZuXEzthDd6QTV6zOjrN3erMA/edit?usp=sharing&ouid=113847654012447316727&rtpof=true&sd=true

References

[1] Roustaei, N. (2024). Application and interpretation of linear-regression analysis. *Medical Hypothesis, Discovery and Innovation in Ophthalmology*, *13*(3), 151.

[2] Uyanık, G. K., & Güler, N. (2013). A study on multiple linear regression analysis. *Procedia-Social and Behavioral Sciences*, *106*, 234-240.

[3] Zhou, Z., Qiu, C., & Zhang, Y. (2023). A comparative analysis of linear regression, neural networks and random forest regression for predicting air ozone employing soft sensor models. *Scientific Reports*, *13*(1), 22420.

[4] Mullahy, J., & Norton, E. C. (2024). Why transform Y? The pitfalls of transformed regressions with a mass at zero. *Oxford Bulletin of Economics and Statistics*, *86*(2), 417-447.